SMART POWER MANAGEMENT SYSTEM

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Abstract This paper presents the design and hardware implementation of an AI-integrated Smart Power Management System (SPMS) tailored for Velavan Hypermarket, a 5-story commercial facility in Thoothukudi. The facility currently faces challenges with a connected load of over 650 KW exceeding its sanctioned 600 KW Maximum Demand (MD) limit, leading to potential utility penalties and high energy consumption of approximately 6000 units per month. To address this, the proposed system utilizes an ESP32 microcontroller integrated with the Blynk IoT platform and a Machine Learning predictive layer to execute three specific energy optimization algorithms: (1) AI-Predictive Load Shifting to proactively isolate non-essential Air Handling Units (AHUs) before peak demand breaches, (2) Zonal Daylight Harvesting to switch off 60% of lighting in the furniture display zones, and (3) Grid-Tied Solar PV integration. The hardware prototype and AI logic demonstrate real-time monitoring and automated predictive load control, effectively reducing grid dependency and ensuring strict compliance with the sanctioned demand limit..

Keywords— Smart Power Management System (SPMS), Machine Learning, Predictive Load Shedding, Maximum Demand Control, Daylight Harvesting, Grid-Tied Solar PV.

I. INTRODUCTION

The Commercial buildings are significant consumers of electrical energy, accounting for a substantial portion of the global power demand. In large-scale retail environments, energy is primarily consumed by HVAC systems and lighting. Inefficient management of these loads often leads to exceeding the sanctioned Maximum Demand (MD) set by utility providers.

This paper focuses on a case study of Velavan Hypermarket Pvt Ltd in Thoothukudi. The facility operates on a 22KV High Tension (HT) supply stepped down to 440V via a 1250 KVA transformer. The current infrastructure supports a massive connected load exceeding 650 KW, distributed across five distinct operational floors:

- **1st Floor:** Textile, Supermarket, Jewellery (Gold), and KFC.
- **2nd Floor:** Home Appliances and Mobile Section.
- **3rd Floor:** Stationary and Fancy Items.
- **4th Floor:** Living Area Furniture and Electronics (TV/AC).
- **5th Floor:** Gaming Zone, Food Court, and Bedroom Furniture (Cots/Bureaus).

Data analysis reveals that simultaneous operation of HVAC systems across all 5 floors, particularly the high-intensity spot lighting required for the furniture displays on the 5th floor and the gaming zone load, often pushes the demand beyond the sanctioned 600 KW limit during peak hours, incurring severe penalties.

II. SYSTEM ANALYSIS

A. Existing System Limitations The current power management infrastructure relies entirely on manual operation.

1. **Manual Latency:** Maintenance staff cannot react instantly to load spikes caused by the 5th-floor gaming zone or 1st-floor crowds, leading to MD breaches.
2. **Energy Waste:** Lighting on the 4th (Living Furniture) and 5th floors (Bedroom Furniture) remains ON at 100% capacity even during peak daylight hours, despite large window frontages.
3. **Grid Dependency:** The facility relies 100% on the TANGEDCO grid for its 6000-unit monthly consumption, ignoring rooftop solar potential.

B. Proposed Smart System The proposed SPMS introduces a closed-loop Artificial Intelligence of Things (AIoT) control architecture:

1. **Automation & AI:** An ESP32 microcontroller continuously monitors load parameters while an ML inference model analyzes usage trends.

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2. **Predictive Load Shedding:** Instead of waiting for the load to breach the threshold, the AI predicts impending spikes. If the trajectory points to a breach, the system proactively isolates non-essential AHUs on the 3rd and 4th floors (Low Priority Zones) to protect the 1st floor (High Priority).
3. **Hybrid Energy Source:** Integration of Grid-Tied Solar PV to offset daytime loads.

III. PROPOSED METHODOLOGY

The SPMS executes three distinct algorithms running simultaneously, optimized by the ML layer.

A. Algorithm 1: AI-Predictive Load Shifting

The system monitors Total Grid Power P_{Total} and incorporates historical time-of-day features. The total power is calculated as:

$$P_{Total} = P_{Chiller} + \sum_{i=1}^{n} P_{AHU(i)} + P_{light}$$

Instead of a reactive limit, an ML model forecasts the load. If the predicted $P_{Total} \geq 600$ KW the system triggers relays

Priority Logic: The system targets the 3rd Floor (Stationary) and 4th Floor (Living Furniture) for load shedding. These zones contain non-perishable goods, meaning a 15-minute HVAC cutout does not impact business operations, unlike the 1st Floor (Perishables/Gold) or 5th Floor (Dining/Gaming). B.

B. Algorithm 2: Zonal Daylight Harvesting The 4th Floor and 5th Floor feature extensive windows to display products. Light Dependent Resistors (LDRs) measure ambient sunlight (Ev).

$$P_{Light} = \{0.4\ P_{Rated},\ \text{if } E_v > 500\ lx,$$

$$P_{Light} = \{P_{Rated},\ \text{if } E_v \leq 500\ lx$$

If ambient light exceeds 500 lx, the system switches OFF 60% of the lighting circuits on the 4th and 5th floors, leaving only essential aisle lighting active.

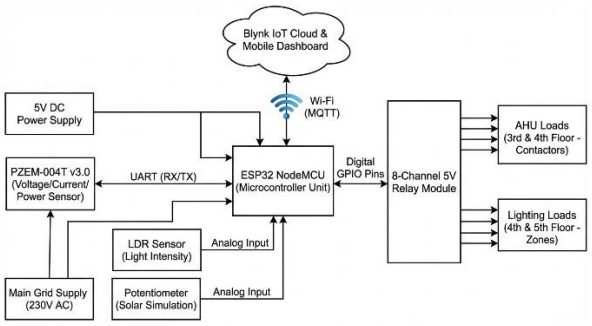
C. Algorithm 3: Grid-Tied Solar PV Integration

To reduce the 6000-unit monthly consumption, the system integrates a Solar PV input (P_{Solar})

The net grid power P_{Net} drawn is minimized:

$$P_{net} = P_{Total} - P_{solar}$$

IV. Block Diagram



V.HARDWARE AND SOFTWARE IMPLEMENTATION

A. Hardware Architecture

1. **ESP32:** Central controller for logic processing and Wi-Fi communication.
2. **PZEM-004T v3.0:** Measures Voltage, Current, Power (W), and Energy (kWh).
3. **8-Channel Relay Module:**
   - Relay 1-2: Control 3rd Floor AHU (Load Shedding).
   - Relay 3-4: Control 4th Floor AHU (Load Shedding).
   - Relay 5-8: Control Lighting Zones on 4th & 5th Floors.
4. **Sensors:** LDR modules for light detection.

B. Software Platform The firmware is written in C++ using the Arduino IDE. The predictive ML logic is simulated using MATLAB/Simulink. The Blynk Cloud IoT Platform provides the remote user interface, visualizing Grid Power (V0), Solar Power (V1), and Net Energy (V2).

TABLE I.

S.NO	Hardware Specification		
	Component	**Specification**	**Function**
1	ESP32 Node MCU	Dual core WIFI	CPU
2	PZEM-004T V3.0	80-260V AC, 100A	GPM
3	8-Channel Relay Module	5V DC Trigger	Switching Load
4	LDR Sensor	Analog Output	Ambient Light Detection
5	Potentiometer	10k Ohm	Solar PV Simulation
6	Blynk Cloud	IoT Server	Remote Data Visualization

VI. Expected outcome

The design and implementation of the AI-integrated Smart Power Management System (SPMS) provide a predictive and

cost-effective solution for Velavan Hypermarket Pvt Ltd. By transitioning from a manual to an automated AIoT system using the ESP32 Node MCU and Machine Learning forecasting, the facility effectively mitigates the risks of exceeding its sanctioned 600 KW Maximum Demand.

The system's AI logic proactively prioritizing load shedding for the 3rd Floor and 4th Floor AHUs ensures that critical business operations on the 1st and 5th Floors remain uninterrupted, preventing system chattering and sudden blackouts. Furthermore, the integration of Zonal Daylight Harvesting utilizes natural light to cut display lighting costs by approximately 40% during the day. Overall, the prototype validates that intelligent load forecasting combined with Grid-Tied Solar PV integration drastically reduces monthly energy consumption, ensuring strict financial savings and grid stability.

VII. Future Scope

The successful integration of Machine Learning for predictive load shedding, the SPMS establishes a robust AIoT foundation. Future enhancements can expand this intelligent architecture further:

A. Battery Energy Storage System (BESS) Integration To maximize the efficiency of the Grid-Tied Solar PV system, a BESS can be introduced. The AI algorithm can be trained to perform energy arbitrage—storing excess solar energy during off-peak morning hours and discharging the batteries during the 5th-floor gaming zone peak periods, further flattening the demand curve.

B. EV Charging Station Dynamic Load Balancing As commercial facilities adopt Electric Vehicle (EV) infrastructure, future iterations of the SPMS can integrate EV charging loads. The predictive controller can dynamically throttle EV charging speeds across the parking lot based on the real-time building load to ensure the 600 KW limit is never breached.

C. Granular Floor-Wise Monitoring Installing individual PZEM-004T sensors for each of the 5 floors would allow the ML model to perform granular energy auditing. This provides hyper-localized data to identify exactly which specific zone contributes the most to demand spikes, improving the accuracy of the forecasting algorithms.

D. Diesel Generator (DG) Synchronization In extreme cases where proactive load shedding is insufficient, the system can be integrated with the facility's Diesel Generator. The ESP32 can send a signal to auto-start the DG and seamlessly transfer the Chiller Plant load before the grid limit is compromised

VIII. Acknowledgment

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